

Round Robin CPU Scheduling Algorithm

Completed

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

Input Format

1. The first line of input contains an integer representing the number of processes.
2. The second line contains an integer representing the Time Quantum.
3. The next lines contain two integers for each process, where
 - The first integer denotes the Arrival Time (AT)
 - The second integer denotes the Burst Time (BT) of the process.

Select Language : C (GCC 9.2.0)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, time_quantum;
5      int at[20], bt[20], remaining_bt[20], ct[20], ta[20];
6      int current_time = 0, completed = 0;
7      float total_tat = 0, total_wt = 0;
8
9      printf("Enter number of processes: \n");
10     scanf("%d", &n);
11     printf("Enter Time Quantum: \n");
12     scanf("%d", &time_quantum);
13
14     for (i = 0; i < n; i++) {
15         printf("Enter AT and BT for P%d: \n", i + 1);
16         scanf("%d %d", &at[i], &bt[i]);
17         remaining_bt[i] = bt[i]; // Keep track of remaining burst time
18     }
19

```

Test Cases

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2